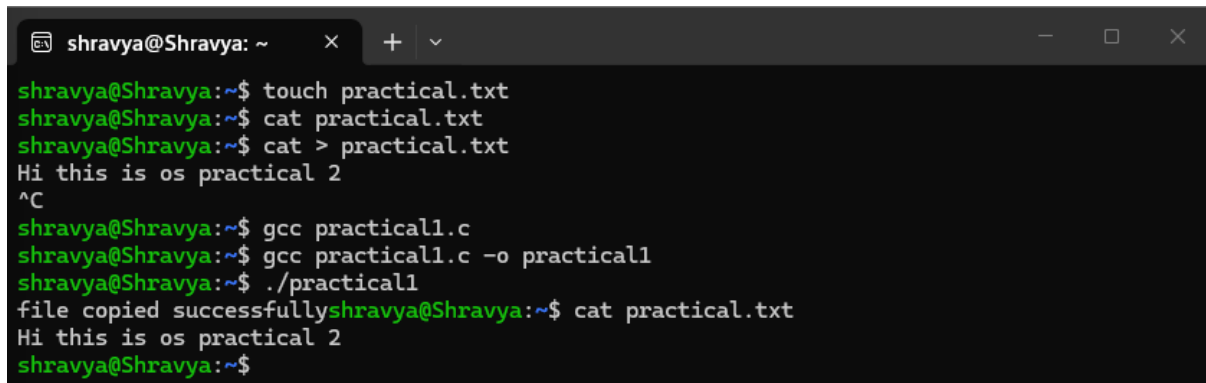


OS PRACTICAL 2

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution. Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

A terminal window with a dark background and light green text. The window title is 'shravya@Shravya: ~'. The terminal shows a series of commands and their outputs. The commands are: 'touch practical.txt', 'cat practical.txt', 'cat > practical.txt', 'gcc practical1.c', 'gcc practical1.c -o practical1', './practical1', and 'cat practical.txt'. The outputs are: 'Hi this is os practical 2' (twice), '^C', and 'file copied successfully'.

```
shravya@Shravya: ~  
shravya@Shravya:~$ touch practical.txt  
shravya@Shravya:~$ cat practical.txt  
shravya@Shravya:~$ cat > practical.txt  
Hi this is os practical 2  
^C  
shravya@Shravya:~$ gcc practical1.c  
shravya@Shravya:~$ gcc practical1.c -o practical1  
shravya@Shravya:~$ ./practical1  
file copied successfullyshravya@Shravya:~$ cat practical.txt  
Hi this is os practical 2  
shravya@Shravya:~$
```

The program uses `open()` to open the source and destination files. `read()` reads the contents of the source file into a buffer, and `write()` writes the contents from the buffer into the destination file. Finally, `close()` closes both files. The output confirms that the file was copied successfully, and the contents of `copy.txt` were verified using `cat`.

```
shravya@Shravya: ~$ strace cat practical.txt
execve("/usr/bin/cat", ["cat", "practical.txt"], 0x7fffc2c942d8 /* 25 vars */) = 0
brk(NULL)                               = 0x5df446011000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x71ce48d14000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=19945, ...}) = 0
mmap(NULL, 19945, PROT_READ, MAP_PRIVATE, 3, 0) = 0x71ce48d0f000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x71ce48cdc000
mmap(0x71ce48ce3000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x71ce48ce3000
mmap(0x71ce48d04000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x71ce48d04000
mmap(0x71ce48d0b000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x71ce48d0b000
mmap(0x71ce48d0d000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x71ce48d0d000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x71ce48cae000
mmap(0x71ce48cb2000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x71ce48cb2000
mmap(0x71ce48cd6000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x71ce48cd6000
mmap(0x71ce48cda000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x71ce48cda000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x71ce48b88000
mmap(0x71ce48b9a000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x71ce48b9a000
mmap(0x71ce48c3a000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x71ce48c3a000
mmap(0x71ce48cac000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x71ce48cac000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
```

used the strace utility with the command `cat sample.txt` to observe the system calls generated during execution. The strace output shows that the cat command accesses the file using system calls such as `openat()` and then transfers the file contents to the terminal. The file is finally closed using `close()`. This shows that when a user-space command needs to access a file, it requests the required services from the Linux kernel through system calls.

`open()` → opens sample.txt

`read()` → reads data

`write()` → displays data (conceptually)

`close()` → closes the file